Essential in Mastercam (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**Manufacturing process** Manufacturing concepts Manufacturing types CNC machines NC codes **Activity1:** Study & document various conventional and nonconventional manufacturing process	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	**Mastercam fundamentals** Features of Mastercam GUI of Mastercam Creating geometries Creating 2D profile	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**Creating/Importing 3D models**	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7	**Milling - 2D:** Hole making **Activity2:** Study and document the different parameters control the milling operations in CNC machines	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for practice	Practicals	2.0 Hrs
9	Session 9	**Milling - 2D** Creating counter mill Creating drill Creating pockets Creating area Creating slot mill	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for practice	Practicals	2.0 Hrs
11	Session 11	**Milling - 2D** Creating engrave mill Creating sweep milling Creating revolved cut Creating lofted cut Creating ruled cut	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for practice	Practicals	2.0 Hrs
13	Session 13	**Milling - 2D** Creating dynamic mill Creating dynamic face mill Creating dynamic counter mill **Activity3:** Difference tools used in milling	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for practice	Practicals	2.0 Hrs
15	Session 15	**Milling 3D** **Roughing** Optirough Pocket Area rough	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for practice	Practicals	2.0 Hrs
17	Session 17	**Milling 3D** **Finishing** Finishing using waterline Finishing using scallop	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for practice	Practicals	2.0 Hrs
19	Session 19	**Milling 3D** Hybrid finishing Raster finishing	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for practice	Practicals	2.0 Hrs